Introduction to Machine Learning Methods

Reinforcement Learning

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Outline

- Passive learning
- Active learning

Reference book: Chapters 21

Reinforcement learning – concept

- Learn how to behave successfully to achieve a goal while interacting with an external environment
- Learn from experiences.
- Unlike classification there are no examples of correct answers: try to learn.

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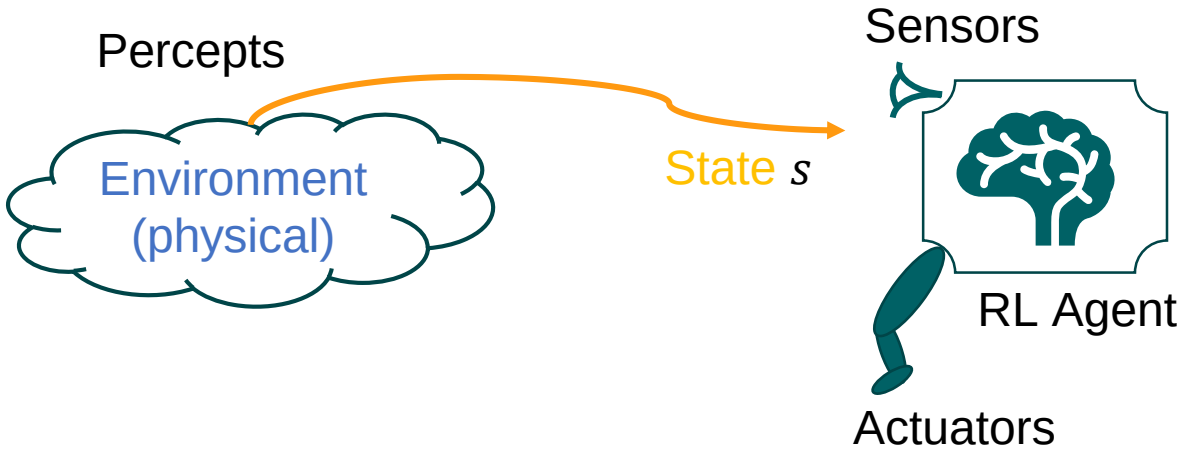

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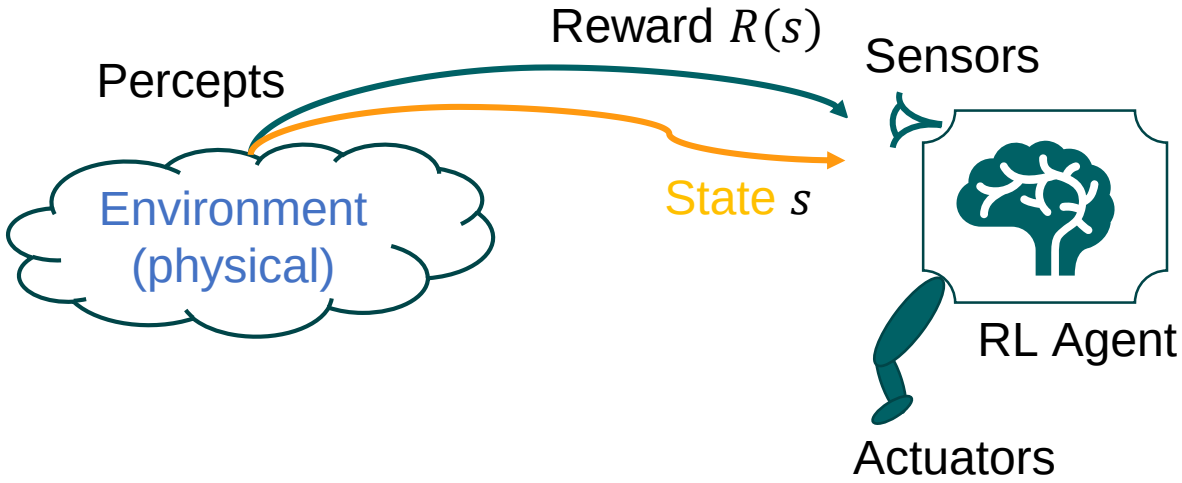

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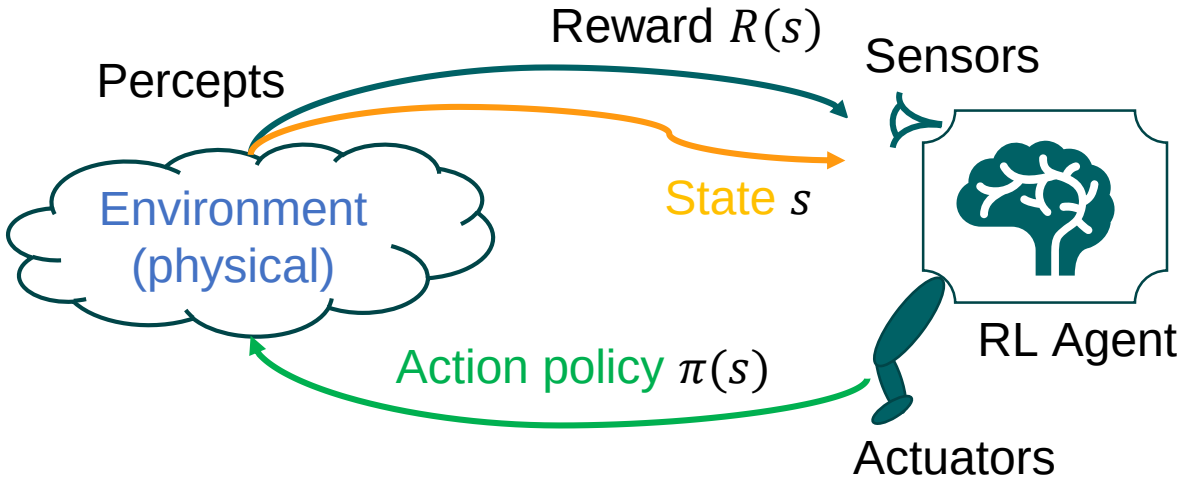

RL model

- Reinforcement:
  - Change the state and trigger rewards or punishments (i.e., negative rewards)
  - Each percept is enough to determine the state (the state is accessible)
  - ***Note*:** The agent can decompose the reward component from a percept.

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  - Maximize cumulative reward during an episode.
- Exploration-Exploitation **trade-off**
  - Exploration: Get exposed to the environment.
  - Exploitation: Utilize knowledge

Agent types

- Utility-based agents:
  - Learn a utility function of states uses it to select actions to maximize the expected outcome utility.
- Q-learning agents:
  - Learns the expected utility of taking a particular action a in a particular state s (Q-value of the pair (s, a)).
- Reflex agents:
  - Learn a policy that maps directly from states to actions

Passive Learning

Assumptions

- The agent's policy is fixed – in a state s it always executes the action $a = \pi(s)$
- Transition model $P(s'|s, a)$ is not known.
- Reward function for **each** state $R(s)$ is unknown

State-based representation in a fully observable environment

- Goal: Learn the utilities function $U^{\pi}(s)$ (or action-state pairs)

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- Example:

Optimal policy for a 4x3 example,
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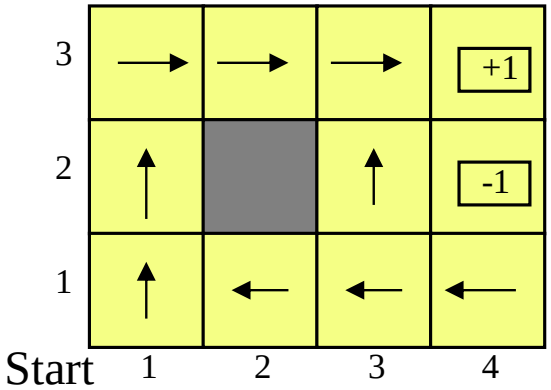

The utilities of states

0.812	0.868	0.912	+1
0.762		0.660	-1
0.705	0.655	0.611	0.388
1	2	3	4

Passive learning task problem

- The passive learning task is equivalent to the **policy evaluation** task of the **policy iteration** algorithm
- The main difference is that the passive learning agent does not know the **transition model** $P(s'|s,a)$
- Reward function for **each** state $R(s)$ is unknown

Solution?

Solution to passive learning task

- Step 1:
  - The agent executes a set of **trials** in the environment using its policy π

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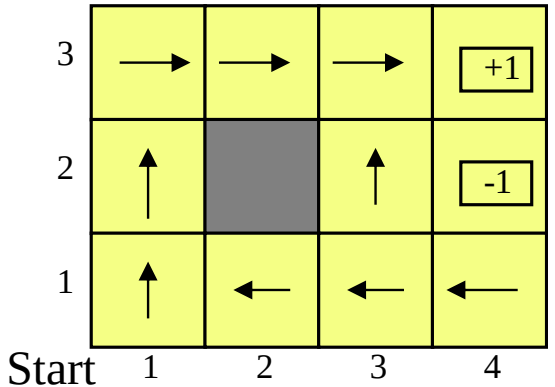

Solution to passive learning task

- Step 1:
  - The agent executes a set of **trials** in the environment using its policy π
- Step 2
  - In each trial, the agent starts in state (1,1) and experiences a sequence of state transitions until it reaches one of the terminal states, (4,2) or (4,3).
  - Its percepts supply both the current state and the reward received in that state.

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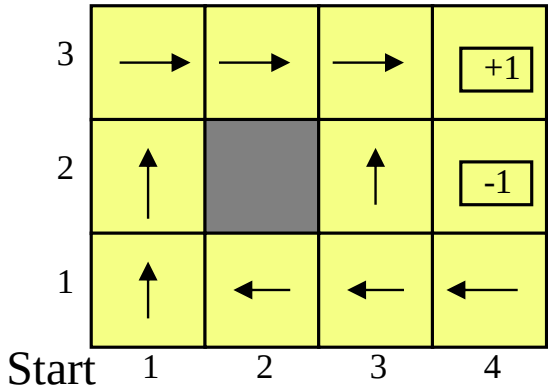
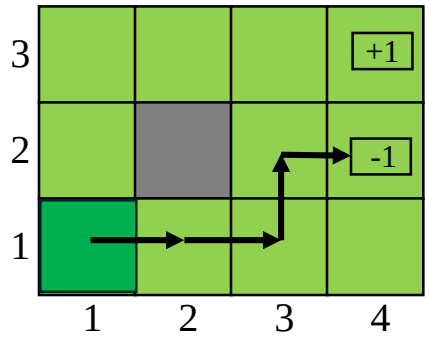


Example of different trials

Trial 1:

$(1,1)_{-0.04} \rightsquigarrow (2,1)_{-0.04} \rightsquigarrow (3,1)_{-0.04} \rightsquigarrow (3,2)_{-0.04} \rightsquigarrow (4,2)_{-1}$

Here, each percept of a state is associated with the reward received



Example of different trials

Trial 2:

$$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$$

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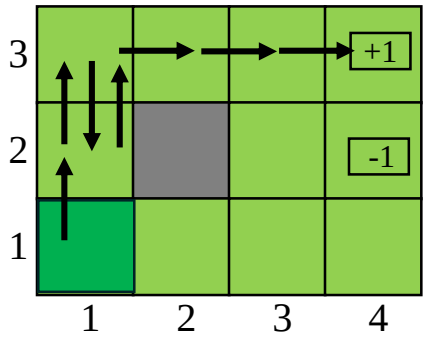

Example of different trials

Trial 3:

$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (3,2)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$

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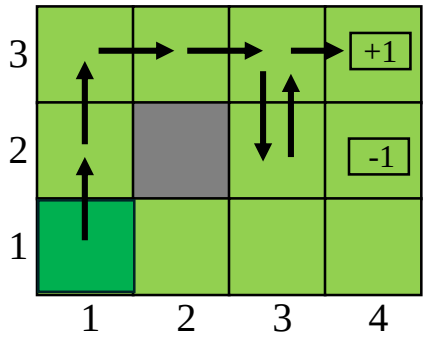


Utility computation

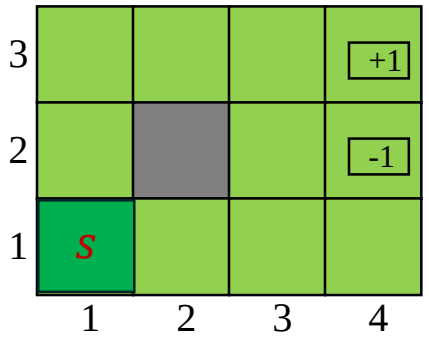
- The goal is to use the information about rewards to learn the utility $U^{\pi}(s)$ associated with each nonterminal state s given by the expected sum of discounted rewards obtained if policy π is executed:

$$U^{\pi}(s = s_0) = E\left[\sum_{t=0}^{\infty} \gamma^t R(s_t)\right]$$

Direct utility estimation

- The main idea is that the utility $U^{\pi}(s)$ of a state s is the expected total reward from s onward (called the expected **reward-to-go**), while each ***trial*** sequence provides a *sample* of this quantity for each state visited by policy π!

- Let's see our examples for $\gamma = 1$:



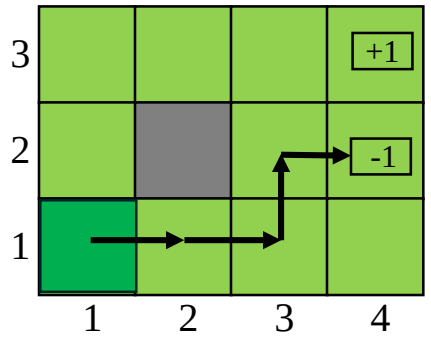
Example

- For the **first** trial:

$$(1,1)_{-0.04} \rightsquigarrow (2,1)_{-0.04} \rightsquigarrow (3,1)_{-0.04} \rightsquigarrow (3,2)_{-0.04} \rightsquigarrow (4,2)_{-1}$$

the expected utility is given by

$$U^{\pi}(1,1) = E\left[\sum_{t=0}^{\infty} \gamma^t R(s_t)\right]_{\gamma=1} = 4 \times (-0.04) + (-1) = -1.16$$



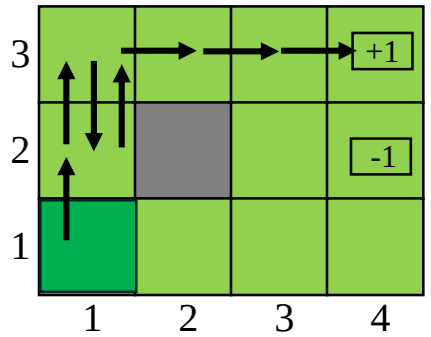
Example

- For the **second** trial:

$$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$$

the expected utility for $s = (1,1)$ is given by

$$U^{\pi}(1,1) = E\left[\sum_{t=0}^{\infty} \gamma^t R(s_t)\right]_{\gamma=1} = 7 \times (-0.04) + 1 = 0.72$$



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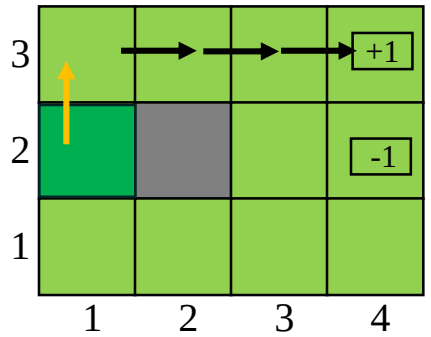
$$U^{\pi}(1,2) = 4 \times (-0.04) + 1 = 0.84$$



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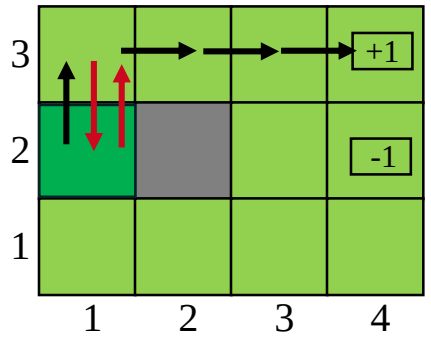
$$U^{\pi}(1,2) = 6 \times (-0.04) + 1 = 0.76$$



Example

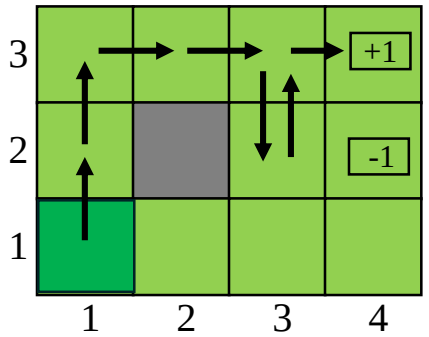
- For the **third** trial:

$$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (3,2)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$$

the expected utility is given by

$$U^{\pi}(1,1) = E\left[\sum_{t=0}^{\infty} \gamma^t R(s_t)\right]_{\gamma=1} = 4 \times (-0.04) + (-1) = 0.68$$

Remark: this policy provided lower utility in the first sequence!



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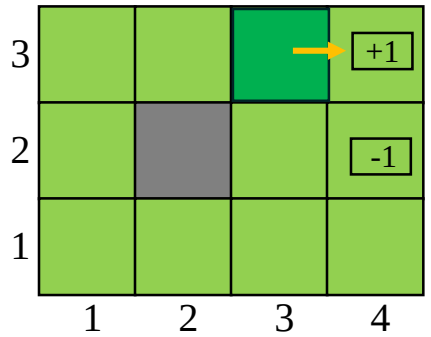
- For $s = (3,3)$ we have two values:

$$U^{\pi}(3,3) = 1 \times (-0.04) + 1 = 0.96$$



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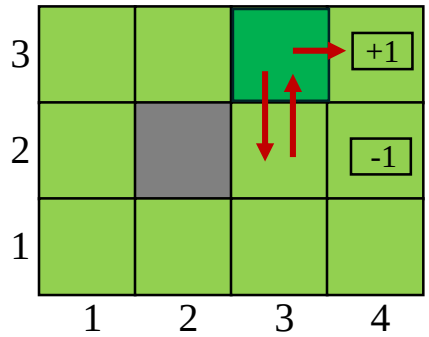
$$U^{\pi}(3,3) = 3 \times (-0.04) + 1 = 0.88$$



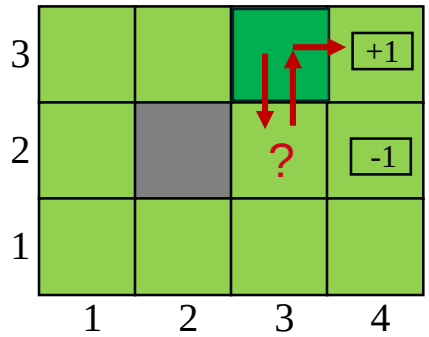
Dependence between the states

- For $s = (3,3)$ we have two values:

$$U^{\pi}(3,3) = 1 \times (-0.04) + 1 = 0.96$$
$$U^{\pi}(3,3) = 3 \times (-0.04) + 1 = 0.88$$

- What is the utility of $s = (3,2)$?
  - *Hint*: use Bellman equation!

$$U^{\pi}(s) = R(s) + \gamma \sum_{s'} P(s'|s, \pi(s)) U^{\pi}(s')$$

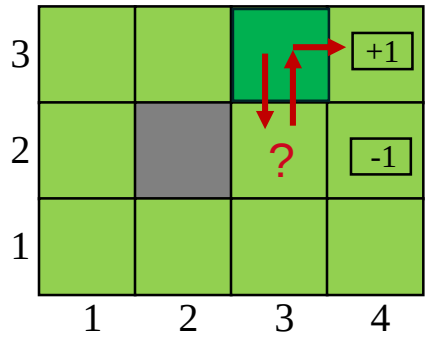


Remark

- Remember that the utilities of states are not independent!

$$U^{\pi}(s) = R(s) + \gamma \sum_{s'} P(s'|s, \pi(s)) U^{\pi}(s')$$

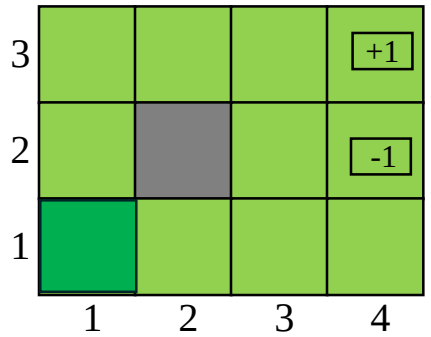
Logic: Since state (3,2) leads to a high utility state (3,3) it is also likely to have high utility!



Direct utility estimation algorithm

- At the end of each sequence, we compute the following
  - Observed **reward-to-go** for each state
  - Update the estimated utility for that state accordingly by a running average for each state in a table.
- In the limit of infinitely many trials, the sample average will converge to the true expectation

$$E\left[\sum_{t=0}^{\infty} \gamma^t R(s_t)\right]$$



Remark

- Direct utility estimation is just a sample of supervised learning
  - Each example has the **state** as input and the observed **reward-to-go** as output.

Adaptive dynamic programming

ADP agent

- For a passive learning agent, an **adaptive dynamic programming** (or ADP) agent solves the corresponding Markov decision process using a dynamic programming method!

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- ADP agents computes the utilities of the states by plugging the learned transition model $P(s'|s,\pi(s))$ and the observed rewards $R(s)$ into the Bellman equations
- The utilities it needs to learn are those defined by
- the *optimal* policy;

linear (no max operator)

ADP agent

- For a passive learning agent, an **adaptive dynamic programming** (or ADP) agent solves the corresponding Markov decision process using a dynamic programming method!
- ADP agents computes the utilities of the states by plugging the learned transition model $P(s'|s,\pi(s))$ and the observed rewards $R(s)$ into the Bellman equations

$$U^{\pi}(s) = R(s) + \gamma \sum_{s'} P(s'|s,\pi(s))U^{\pi}(s')$$

linear (no max operator)

- **Solution**: see the Policy iteration (**modified policy iteration)** algorithm in the previous lecture!!!

$$U_{i+1}(s) \leftarrow R(s) + \gamma \sum_{s'} P(s'|s,\pi_i(s))U_i(s')$$

Learning principle

- The process of learning the model is easy because the environment is fully observable
  - This means that we have a supervised learning task where the input is a state–action pair and the output is the resulting state
  - Here, the transition model is represented by a table of probabilities $P(s'|s, \pi(s))$
- **Goal**:
  - Keep a track of how often each action outcome occurs and estimate the transition probability $P(s'|s, a = \pi(s))$ from the frequency with which s' is reached when executing $a = \pi(s)$ in s

Example

- For the trial sequences:

$$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (3,2)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$$

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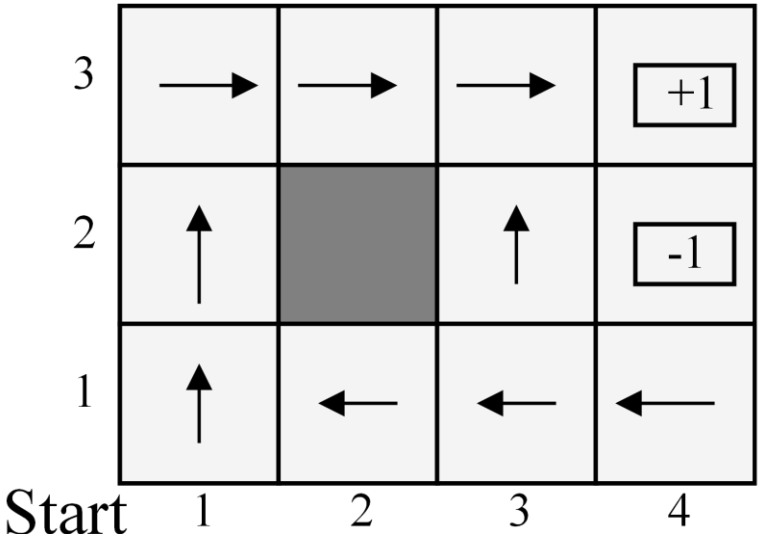
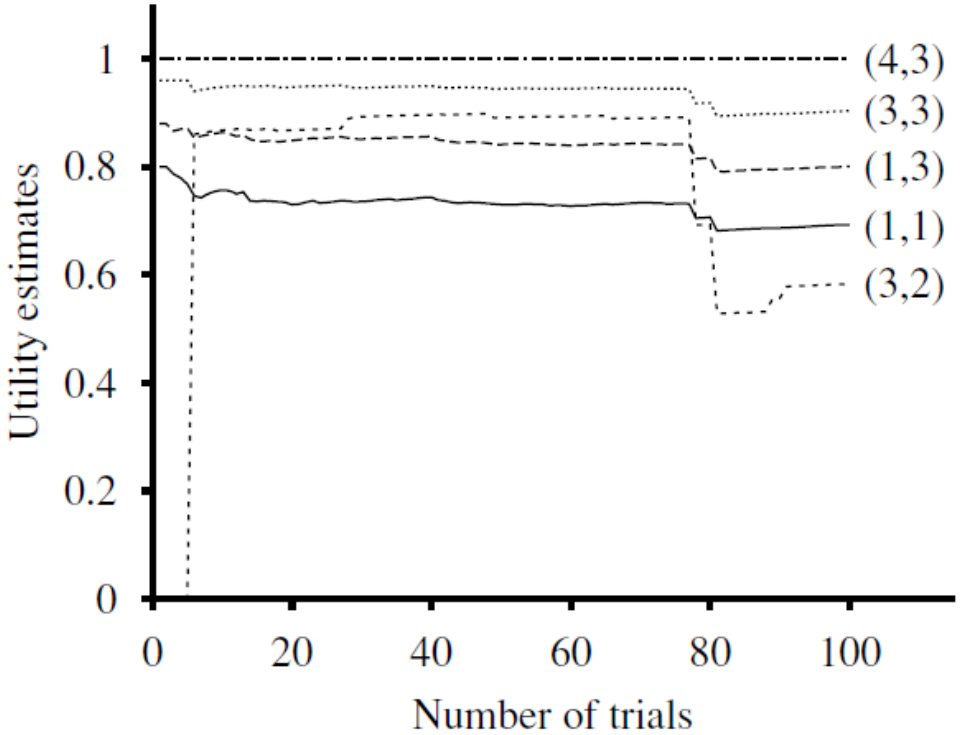
- The action "$a = Right$" is executed three times in $s = (1,3)$ and two out of three times the resulting state is $s' = (2,3)$

$$P((2,3)|(1,3), a = Right) = \frac{2}{3}$$

The passive ADP learning curves for the 4×3 world

Optimal policy for a 4x3 example, with $\gamma = 1$ and $R(s) = -0.04$





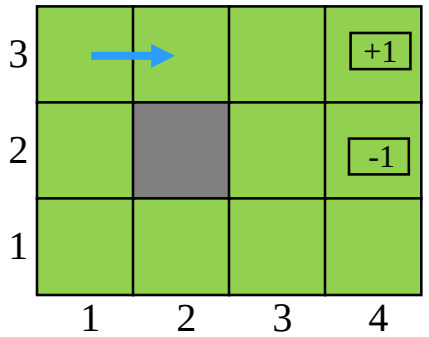
Remarks

- ADP agent is limited by its ability to learn the transition model
- ADP is a standard for comparison of other reinforcement learning algorithms
- However, ADP is intractable for large state spaces.

Temporal-difference learning

Use observed transitions to adjust the utilities

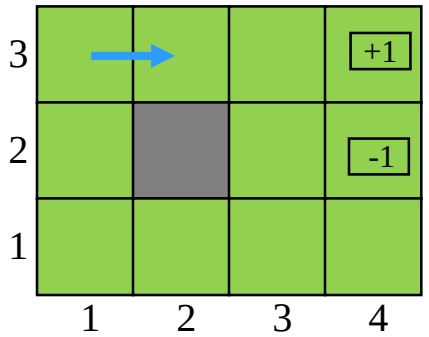
- TD: Adjust the estimated utility value of the current state based on its immediate reward and the estimated value of the next state.



Use observed transitions to adjust the utilities

- TD: Adjust the estimated utility value of the current state based on its immediate reward and the estimated value of the next state.
- Let's check the third trial sequence:

$$(1,1)_{-0.04} \mapsto (1,2)_{-0.04} \mapsto (1,3)_{-0.04} \mapsto (2,3)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (3,2)_{-0.04} \mapsto (3,3)_{-0.04} \mapsto (4,3)_{+1}$$



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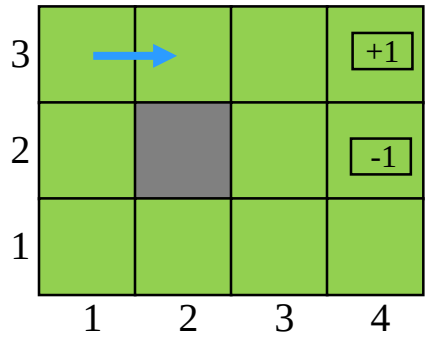

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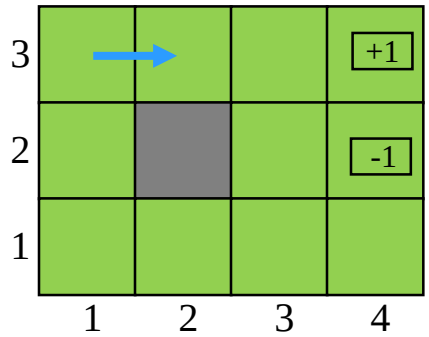
- Assume in the second trial we estimated $U^{\pi}(1,3) = 0.84$ and $U^{\pi}(2,3) = 0.92$
- Thus, for transition in blue we estimate the utility:

$$U^{\pi}(1,3) = -0.04 + U^{\pi}(2,3) = 0.88$$

Hence, the current estimate of $U^{\pi}(1,3) = 0.84$ is low and should be updated



Temporal difference method

- TD: Adjust the estimated utility value of the current state based on its immediate reward and the estimated value of the next state.

- When a transition occurs from state s' to state s, the updating rule is given by

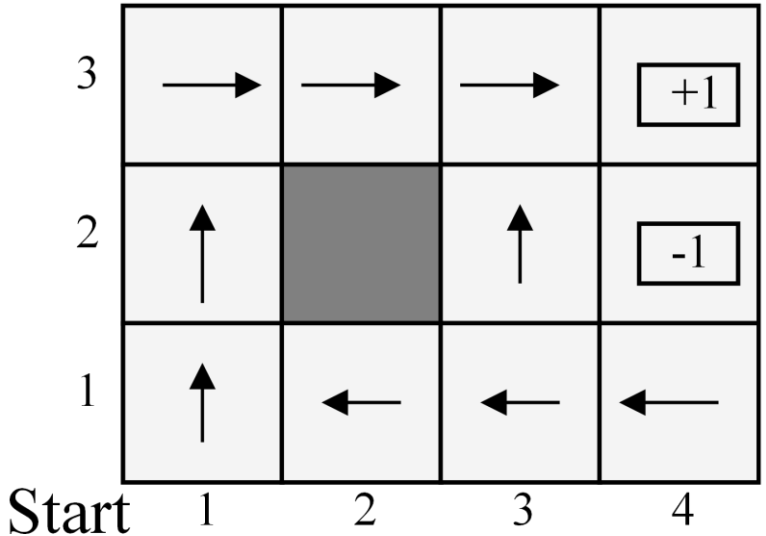
$$U^{\pi}(s) \leftarrow U^{\pi}(s) + \alpha[R(s) + \gamma U^{\pi}(s') - U^{\pi}(s)]$$

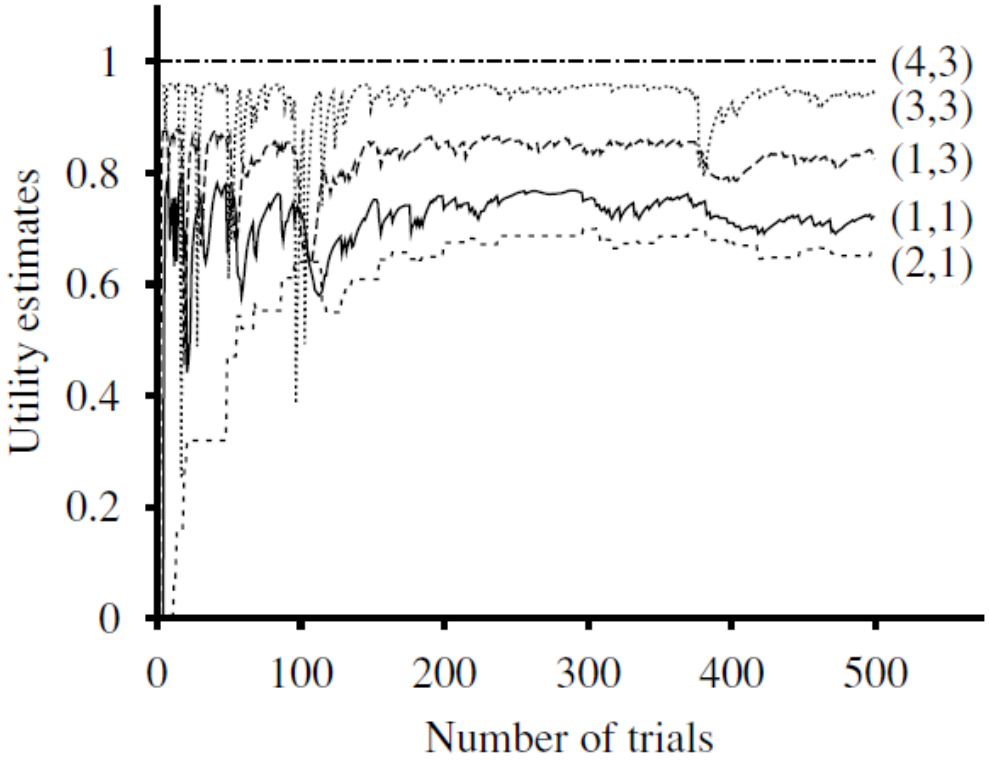
Difference in utilities between successive states; temporal difference

  - α is the learning rate parameter

The TD learning curves for the 4 × 3 world

Optimal policy for a 4x3 example, with $\gamma = 1$ and $R(s) = -0.04$





Active Learning

Notes

- A passive learning agent has a fixed policy π that determines its behavior
- An active agent must ***decide*** what ***actions*** to take (from *experiences*)
  - What their outcomes may be (both on learning and receiving the rewards in the long run)?

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- An active agent must ***decide*** what ***actions*** to take (from *experiences*)
  - What their outcomes may be (both on learning and receiving the rewards in the long run)?

- Agent learns a complete model with outcome probabilities for all actions, rather than just the model for the fixed policy
  - Passive ADP agent, and
  - Considering that the agent has a choice of actions
- How to do it?

Active learning formulation

- The utilities it needs to learn are those defined by the *optimal* policy by Bellman equations:

$$U(s) = R(s) + \gamma \max_{\boldsymbol{a}} \sum_{s'} P(s'|s, \boldsymbol{a}) U(s')$$

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  - Alternatively, if it uses policy iteration, the optimal policy is already available, so it should simply execute the action the optimal policy recommends

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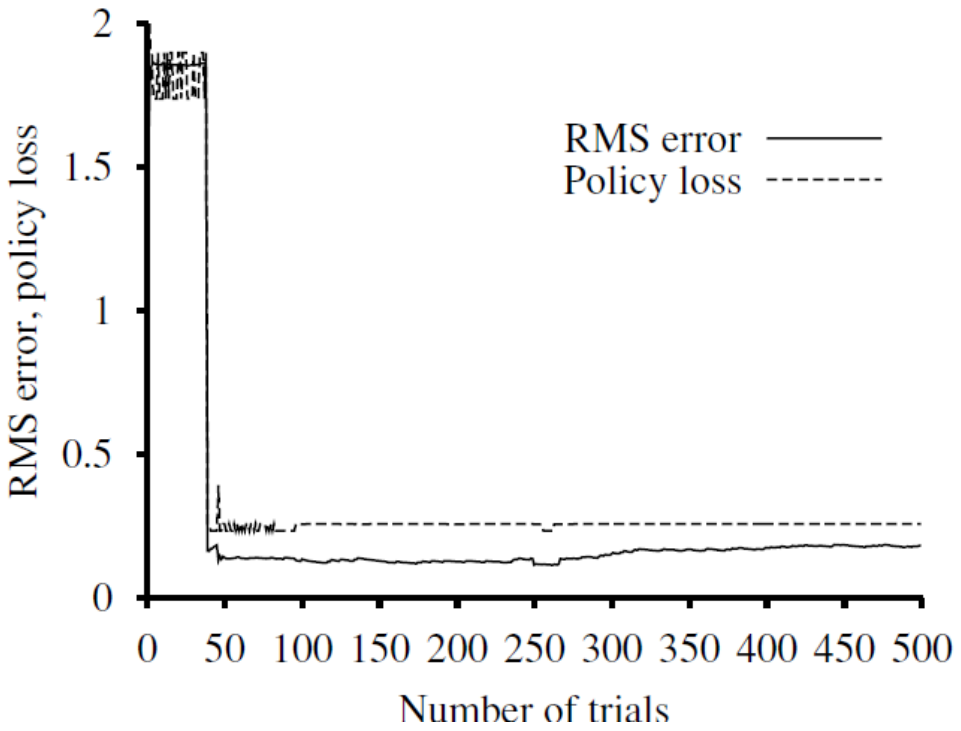
But can the agent learn the true utilities or the true optimal policy!

should simply execute the action the optimal policy recommends

Greedy agent

- In the 39th trial, it finds a policy that reaches the +1 reward along the lower route via **(2,1), (3,1), (3,2), and (3,3)**
- After experimenting with minor variations, from the 276th trial onward, it sticks with this policy, never learning the utilities of the other states and never finding the optimal route via **(1,2), (1,3), and (2,3)**
  - **Greedy agent**.



Remarks

- How can it be that choosing the optimal action leads to suboptimal results?
  - The answer is that the learned model is not the same as the true environment; what is optimal in the learned model can therefore be suboptimal in the true environment
- Unfortunately, the agent does not know what the true environment is, so it cannot compute the optimal action for the true environment
  - What, then, is to be done?
- What the greedy agent has overlooked is that actions do more than provide rewards according to the current learned model
  - Actions also contribute to learning the **true model** by affecting the percepts that are received.

Exploitation vs Exploration

- By improving the model, the agent will receive greater rewards in the future
- An agent therefore must make a tradeoff between **exploitation** to maximize its reward—as reflected in its current utility estimates—and **exploration** to maximize its long-term well-being

- Since the learned model is not a copy of the real environment, the agent needs to trade off between exploration and exploitation:
  - Exploration: With a probability, ε, the agent selects the action randomly, maximizing the agent's long-term performance
  - Exploitation: With a probability, $1 - \varepsilon$, the agent selects the best known action, which is based on its policy.

Learning an action-utility function

Let’s construct an active temporal-difference learning agent

Note

- Now, the agent is no longer equipped with a fixed policy
  - To learn a utility function U, it will need to learn a model to be able to choose an action based on U via one-step look-ahead.
- Q-learning…

Q-learning principle

- **Q-learning**, which learns an action-utility representation $Q(s, a)$ instead of learning utilities
  - Model free learning
  - $Q(s, a)$ denotes the value of doing action *a* in state *s*

Q-learning principle

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- Q-functions may seem like just another way of storing utility information, but they have a very important property

Model-free learning

- Given an estimated model, the update equation for Q-values is given by

$$Q(s,a) = R(s) + \gamma \sum_{s'} P(s'|s,a) \max_{a'} Q(s',a')$$

- This does, however, require that a model also be learned, because the equation uses $P(s'|s,a)$

- Now, we introduce TD agent that learns a Q-function without the need of a model $P(s'|s,a)$, either for learning or for action selection

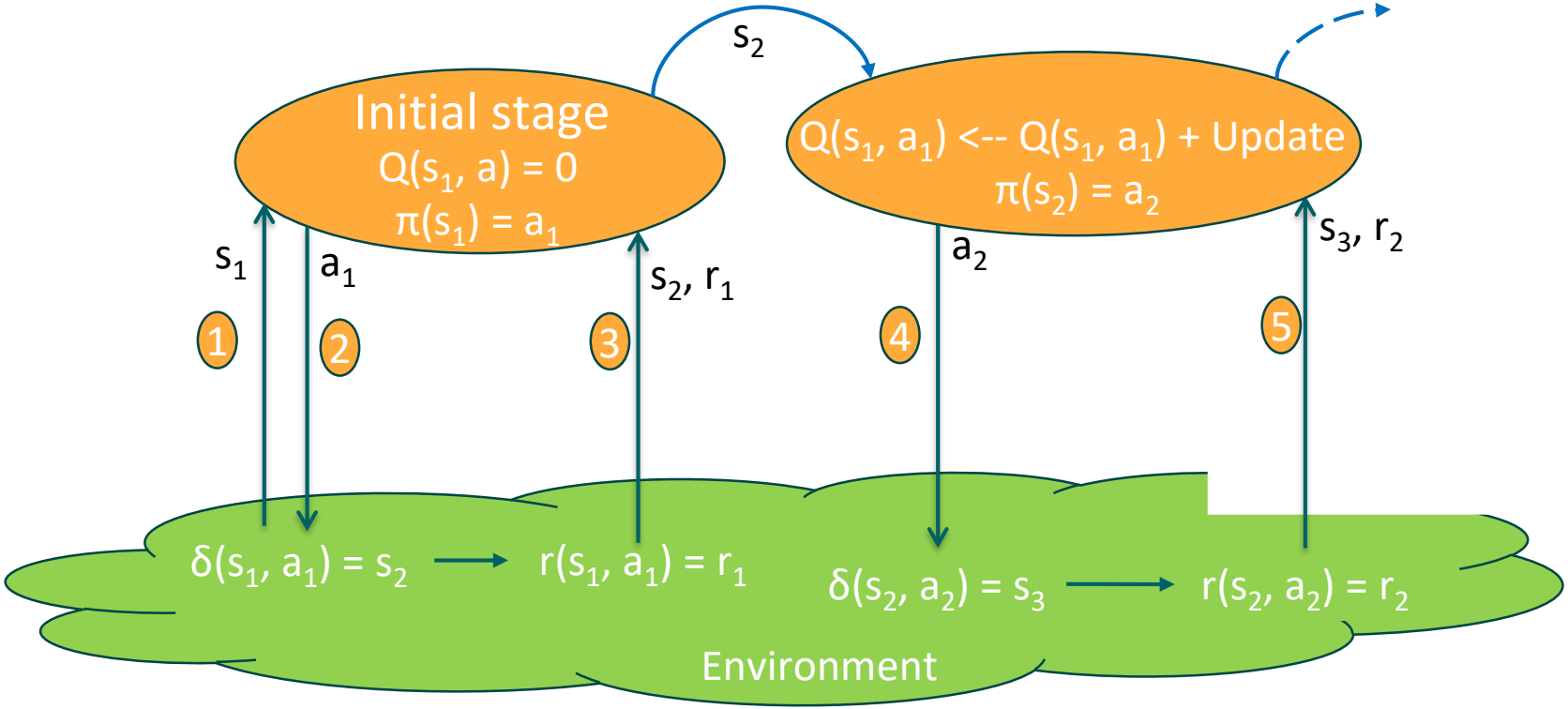
TD Q-learning

- The update equation for TD-Q learning is given by

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[R(s) + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

  - $Q(s,a)$ is calculated whenever action a is executed in state s leading to state s'
  - α is the learning rate

Illustration of Q-value update process



State-Action-Reward-State-Action (SARSA)

- The update equation for TD-Q learning is given by

$$Q(s,a) \leftarrow Q(s,a) + \alpha[R(s) + \gamma Q(s', a') - Q(s,a)]$$

  - Where a' is actual action taken in state s'

- The rule is applied at the end of each s, a, r, s', a' quintuplet—hence the name SARS

Q-learning algorithm principle

1. At *t*=0, initialize *Q*(s, a) for each pair (*s*, *a*)

```
// Adjust the estimated utility value of the current state based on its
immediate reward and the estimated value of the next state.
```

2. Observe the current state *s.*

3. Loop forever {

Select an action *a* with epsilon-greedy algorithm {Random with probability ϵ, and

$a = max_{a'} Q(s', a')$ with probability $1 - \epsilon$}

Receive immediate reward *r* and observe the new state s'.

Update $Q(s, a) \leftarrow Q(s, a) + \alpha[R(s) + \gamma \cdot max_{a'} Q(s', a') - Q(s, a)]$

Update s=s'

}

Remarks

Q-learning: $Q(s,a) \leftarrow Q(s,a) + \alpha \left[R(s) + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$

SARSA: $Q(s,a) \leftarrow Q(s,a) + \alpha [R(s) + \gamma Q(s',a') - Q(s,a)]$

- Q-learning backs up the ***best*** Q-value from the state reached in the observed transition
- SARSA **waits until an action is actually taken** and backs up the Q-value for that action

- A greedy agent always takes the action with best Q-value and the two algorithms are identical.

Thank you!

Chair for Distributed Signal Processing

RWTH AACHEN UNIVERSITY